

Vision Driven Fire Response System

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Vision of the Department

- Creating ethical leaders in the domain of Artificial intelligence and data Science through effectual teaching and learning process to develop emerging technology solutions for the benefits of industry and society with a focus on holistic learning and excellence

Mission of the Department

- Strengthening basic competencies in the domains of Artificial Intelligence and Data Science
- Providing high-quality, value-based technical education and developing technology professionals with creative ideas and compelling leadership abilities.
- Using logical thinking to create and develop cutting-edge products in collaboration with industry stakeholders in order to meet global expectations and requirements
- Enabling graduates to adapt to new technologies via strong fundamentals and lifetime learning

Programme Educational Objectives (PEOs)

- PEO 1: Graduates will demonstrate comprehensive expertise in artificial intelligence, machine learning, and data analysis while applying interdisciplinary knowledge in computer science, statistics, and advanced mathematics to address real-world challenges.
- PEO 2: Graduates will innovate and create impactful solutions ethically, contributing to societal betterment, leveraging their leadership, collaboration, and communication skills for effective teamwork and multidisciplinary endeavors.
- PEO 3: Continuously fostering a culture of lifelong learning, graduates will adapt to evolving technological landscapes, demonstrating adaptability and a commitment to ongoing professional development in the dynamic field of AI and data science.

Programme Outcomes

Engineering Graduates will be able to:

- **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

- **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
- **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

- **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

- **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Programme Specific Outcomes (PSOs)

On the completion of Artificial Intelligence and Data Science program, the students will possess:

- PSO 1: Graduates will be able to apply core principles of science, engineering, and mathematics to develop AI-driven solutions for real-world challenges.
- PSO 2: Graduates will design, develop and implement efficient as well as scalable algorithms using programming languages, data structures and cloud computing technologies.

Programme Specific Outcomes (PSOs)

- PSO 3: Graduates will integrate ethical, social, and security considerations while developing AI-driven intelligent systems, ensuring responsible AI deployment.

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Introduction

- AI-driven Computer Vision for Fire Detection and Safety Automation.
- A system that detects fire using computer vision and dynamically responds using a servo motor.
- Our project enhances fire safety with AI-driven real-time detection, faster alerts, and automated response, reducing false alarms and improving emergency response.

Problem Statement

Our project aims to develop and implement an AI-powered Vision-Driven Fire Response System to create a real-time fire detection solution that enhances safety by reducing false alarms and automating fire suppression in smart buildings and industries.

Objectives

- The Fire Assistant System has four modules: Fire Detection (YOLO detects fire), Communication (PC alerts Arduino), Servo Control (aims water) and Water Pump (extinguishes fire) ensuring efficient fire suppression.
- The Fire Assistant System detects fire with YOLO, alerts Arduino, aims water with a servo, activates a pump, and for quick suppression.
- The Fire Assistant System will automatically detect fire using YOLO, send alerts to Arduino, aim water with a servo motor, and activate the pump to extinguish the fire. This ensures quick response, reduces damage, and enhances fire safety.

Literature Review

Paper Number	Paper Type and Name of publisher	Title,Year,Author	Inferences
One	Conference paper,ResearchGate	Real-time Smoke and Fire Detection Using YOLOv8-based Advanced Computer Vision and Deep Learning ,2023,Shakila Rahman, Syed Muhammad , Hasnat Jamee , Jakaria Khan	Discussed advanced object detection algorithms for fire and smoke.Trained YOLOv8 on diverse fire and smoke datasets.Achieved 82 percent detection accuracy with low false positives.High real-time performance, adaptable to various environments.Limitation was it struggles under low-light conditions.Challenges left was Requires further optimization for nighttime detection.

Literature Review

Paper Number	Paper Type and Name of publisher	Title,Year,Author	Inferences
Two	Journal	Design and Development of 360 Degree Fire Protection System,2022,Dr. Arvind Mahalle , Dr. Nakul Mahalle , Samyak Raisoni , Shubhayu Sarve , Pushkar Madamwar , Sachin Ballewar	Proposed a multi-directional fire detection system with dynamic servo motor response. Simulated multi-directional fire detection to test accuracy and response time. Implemented a sensor-based prototype with servo motors for real-time fire suppression.Successfully demonstrated a 360-degree fire suppression capability, reducing response time and improving precision. Advantages are enhanced efficiency by targeting specific fire sources it is suitable for indoor and outdoor environments.

Literature Review

Paper Number	Paper Type and Name of publisher	Title,Year,Author	Inferences
Three	Journal	Automating Fire Detection and Suppression with Computer Vision: A Multi-Layered Filtering Approach to Enhanced Fire Safety and Rapid Response,2023,Md Safwan Mondal, Nilendu Saha, Saumadeep Guha, Achintya	The paper proposes a multi-layered filtering approach integrating computer vision for fire detection and suppression. Simulations were conducted to evaluate the system's performance in detecting and suppressing fires. Demonstrated enhanced fire detection accuracy and rapid suppression capabilities. Advantages are Improved safety through automated fire response, reducing human intervention.

Literature Review

Paper Number	Paper Type and Name of publisher	Title,Year,Author	Inferences
Four	Journal	A Survey of Vision-based Fire Detection using CNN,2024,Gongsuo ChenTirapot Chandara-supsang (Corresponding author)Xiaodong LuoAnnop TananchanaLei Mu	The paper explores fire detection with convolutional neural networks, highlighting attention mechanisms to reduce false alarms and improve accuracy.Enhanced fire detection accuracy with attention mechanisms and specialized CNN architectures, reducing false alarms in diverse environments.Advantages are enhanced accuracy in detecting fire and distinguishing fire from fire-like objects. Utilizes diverse datasets to improve model generalization.

Literature Review

Paper Number	Paper Type and Name of publisher	Title,Year,Author	Inferences
Five	Journal	Forest Fire Detection System Based on Fuzzy Kalman Filter,2020,Tao Wang, Tingyu Ma, Jianshuo Hu, Jing Song	The proposed system integrates multi-sensor fire detection with a servo-controlled suppression mechanism. Using YOLOv5, it enhances accuracy while minimizing false alarms. The setup includes sensors for fire detection and a servo motor that dynamically directs water for suppression. Simulations test detection accuracy and response time, while hardware integration ensures real-time functionality. Results show improved precision, reduced response time, and

Literature Review

Paper Number	Paper Type and Name of publisher	Title,Year,Author	Inferences
six	Research Article	Wildland Fire Detection and Monitoring Using a Drone-Collected RGB/IR Image Dataset,2022, Bryce Hopkins, Hao Wang, Leo O'Neill, Fatemeh Afghah, Abolfazl Razi, Peter Fulé, Janice Coen, Eric Rowell, Adam	The study demonstrates drones with deep learning as an effective fire detection method, achieving over 78 percentage accuracy in challenging conditions. While it enables real-time, high-resolution monitoring, challenges include dataset dependency, computational limits, and generalization issues across diverse environments. .

Literature Review

Paper Number	Paper Type and Name of publisher	Title,Year,Author	Inferences
Seven	Literature Review	Advancements in Fire Alarm Detection using Computer Vision and Machine Learning,2023, M. Fadli Ridhani, Wayan Firdaus Mahmudy	The paper explores advancements in fire alarm detection using computer vision and machine learning, highlighting their strengths and limitations. It concludes that a hybrid approach combining both methods can improve accuracy, reduce false alarms, and enhance fire detection efficiency in various environments.

Literature Review

Paper Number	Paper Type and Name of publisher	Title,Year,Author	Inferences
Eight	Research article	Fire Detection Using Computer Vision,2024, Akash Yadav, Siddhant Baliga, Ratnesh Upadhyay	The paper discusses fire detection using computer vision and deep learning models like CNN and YOLO for improved accuracy. It highlights early detection benefits while addressing challenges like false alarms and sensor placement.

Gaps identified

- The project's fire detection accuracy may be limited by lack of diverse data, blocked camera views, and slow processing. sample sizes
- Many models struggle with detecting fire in low-light conditions or at night.
- Cost and Infrastructure Limitations

Proposed System

- Dataset collection: We have used a diverse dataset of fire images to ensure the model can learn a wide range of fire patterns.
- Preprocessing: Whether the data is secondary. Perform image preprocessing steps to enhance the quality and variety of the training data.
- Aim: The Fire Assistant System aims to detect and suppress fire using computer vision and automation, ensuring quick response and minimal damage. We used YOLO for fire detection, Arduino for control, a servo motor to aim water, a relay to activate the pump, ensuring quick fire suppression.
- Implementation: The Fire Assistant System uses YOLO for fire detection, alerts Arduino via USB, directs water with a servo, activates a pump.
- Efficiency: The efficiency is evaluated by fire detection accuracy, response time, servo precision, water usage, and alert reliability, ensuring quick and effective fire suppression.

Architectural Diagram

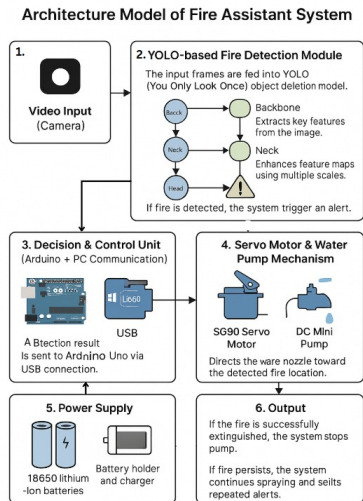


Figure 1: Architecture Diagram

Architectural Workflow

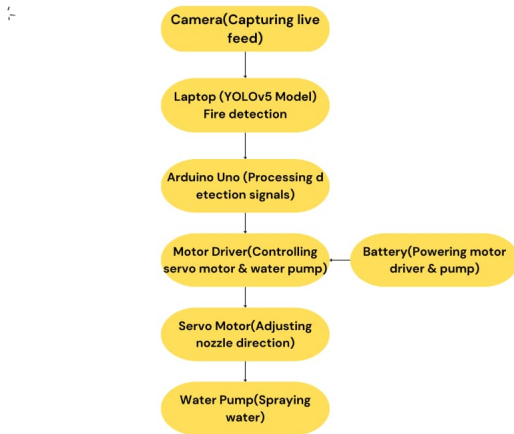


Figure 2: WORK FLOW

Hardware arrangement of proposed system

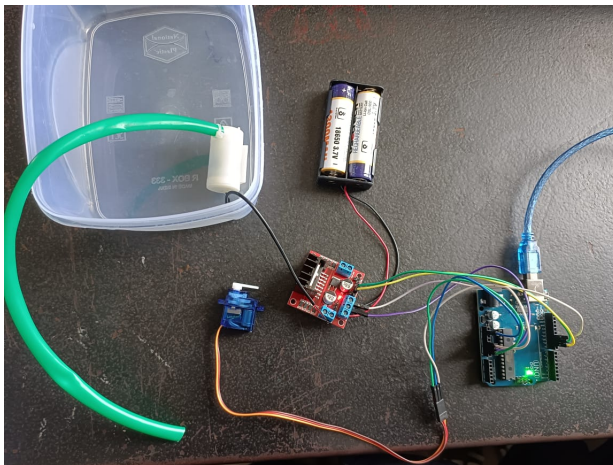


Figure 3: Hardware arrangement

Hardware/Software Requirements

Hardware Requirements

- Arduino Uno – Controls servo motor and water pump.
- Servo Motor – Aims water at the fire.
- Water Pump – Activates to extinguish fire.
- Camera-Captures real-time footage for fire detection.
- Motor Driver – Drives the water pump using battery power.
- Battery - Powering motor driver and pump

Software Requirements

- YOLO (You Only Look Once) – Fire detection using deep learning.
- Languages Used – Python 3.x for coding and system control.
- IDE Used – Arduino IDE for programming Arduino, VS Code/Jupyter Notebook for YOLO and Python scripts.
- Libraries Used – PyTorch

Dataset

Dataset : FIRE DETECTION

Source :The dataset used in this study is obtained from Roboflow.

This dataset consists of 8k images of flames .The total dataset is divided into 80/20 test and training

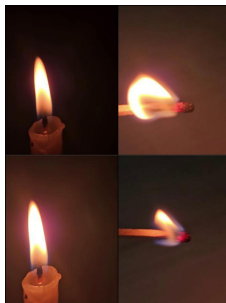


Figure 4: Sample Images

- The Project has been developed, integrating YOLO-based fire detection with real-time camera .
- **Test Accuracy:** The code achieves a test accuracy of 85% , indicating that it correctly classifies fire instances in the test set.

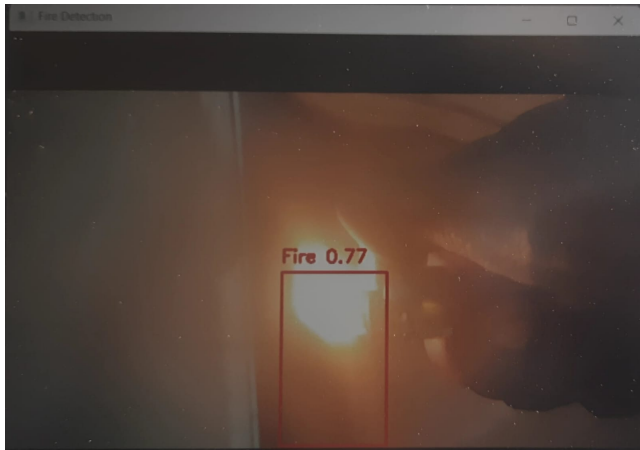
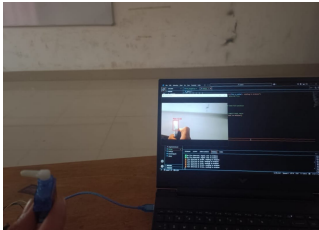
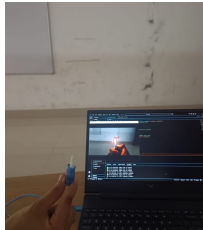


Figure 5: Obtained output

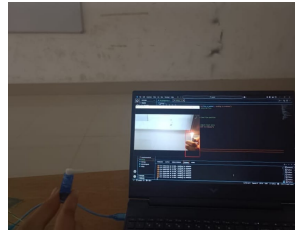
System Response to Fire in Various Location



Fire on Left



Fire in Middle



Fire on Right

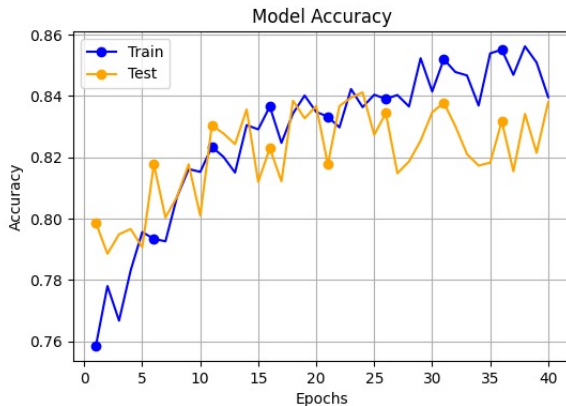
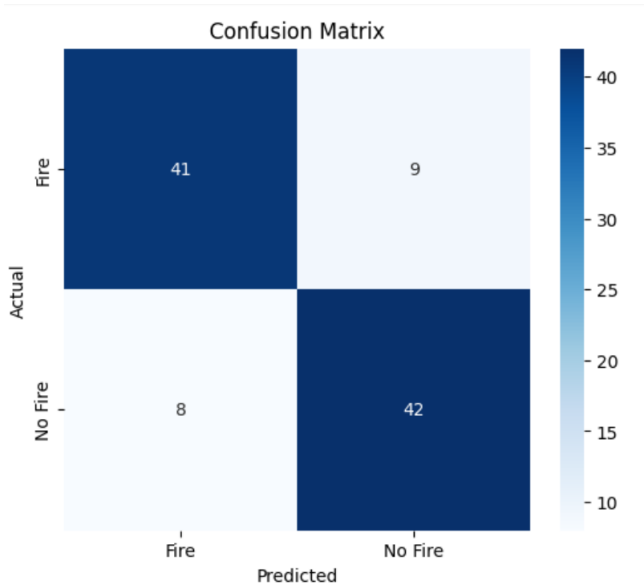


Figure 6: Graphs for Accuracy



Figure 7: Detection per frame

Result



Conclusion

- Successfully implemented a real-time fire detection system using computer vision and a standard mobile camera.
- Integrated Arduino with hardware components (L298 motor driver, pump, and servo motor) for automatic fire response.
- Achieved reliable fire detection without using traditional sensors, relying solely on vision-based techniques.
- Developed a low-cost and scalable prototype suitable for small-scale fire safety applications.
- Ensured quick detection and activation of the extinguishing system to minimize fire-related damage.

FUTURE SCOPE

- Drone-Based Fire Detection – Deploy the system on drones for forest fire detection and rescue operations.
- Integration with IoT – Connect with smart home systems for automated safety measures.
- Wireless Communication – Use Wi-Fi or GSM modules to send alerts remotely instead of USB-based communication.

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THANK YOU